

# Series 2

## Review

1)  $2x(10^{-2})$   
 $20x - 2x^2 = A$   
 Max  $A$  is  $A' = 20 - 4x$   
 $x = 5$   
 $A = 100 - 50 = \underline{50}$

2)  $\sin \theta = 0.5$   
 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

3)  $d(x-2)^2 + (y-1)^2 = 36$

4) 6)

5) a)  $1000 \cdot 2^{0.3 \cdot 24} = 250,000$   
 b)  $N > 10^7$   
 $1000 \cdot 2^{0.36} = 10^7$   
 $2^{0.36} = 10^4$   
 $0.36 = \log_2 10^4$   
 $t = \frac{1321}{0.3} = 44.2$

## Consolidation

6 i)  $\sum_{k=1}^5 (2k+1) = 35$

ii)  $\sum_{k=1}^5 (k^2) = 30$

7 i)  $\sum_{k=1}^6 (2k^2 - 1) = 2(1) - 1 + 2(4) - 1 + 2(9) - 1 + 2(16) - 1 + 2(25) - 1 + 2(36) - 1$   
 $= 1 + 7 + 17 + 31 + 49 + 71$   
 $= 176$

ii)  $\sum_{k=1}^3 k^2 = \sum_{k=0}^3 \frac{k}{k+1}$   
 $= (1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}) - (\frac{0}{1} + \frac{0}{2} + \frac{0}{3} + \frac{0}{4})$   
 $= \frac{15}{12}$

iii)  $\sum_{k=0}^4 (2k+1) - \sum_{k=0}^4 (2k-1)$   
 $2(0+1) + 2(2) + 2(3) + 2(4) + 2(5) - (2(0-1) + 2(1) + 2(2) + 2(3) + 2(4))$   
 $= 10$

## Exam Questions

1 a)  $a_1 = 3$   
 $a_2 = 9$   
 $a_3 = 3$

ii) period 2

2 a)  $u_1 + 2u_2 + u_3$

$2 + 2(k-12) + k - \frac{24}{k-12} = 0$

$2(k-12) + 2(k-12)^2 + k(k-12) - 24 = 0$

$2k - 24 + 2k^2 - 48k + 288 + k^2 - 12k - 24 = 0$

$3k^2 - 58k + 240 = 0$

$(3k - 40)(k - 6) = 0$

$k = 6$  ( $k \neq \frac{40}{3}$  since  $k$  is an integer)